

1. Out of Boundary Paths

Aim

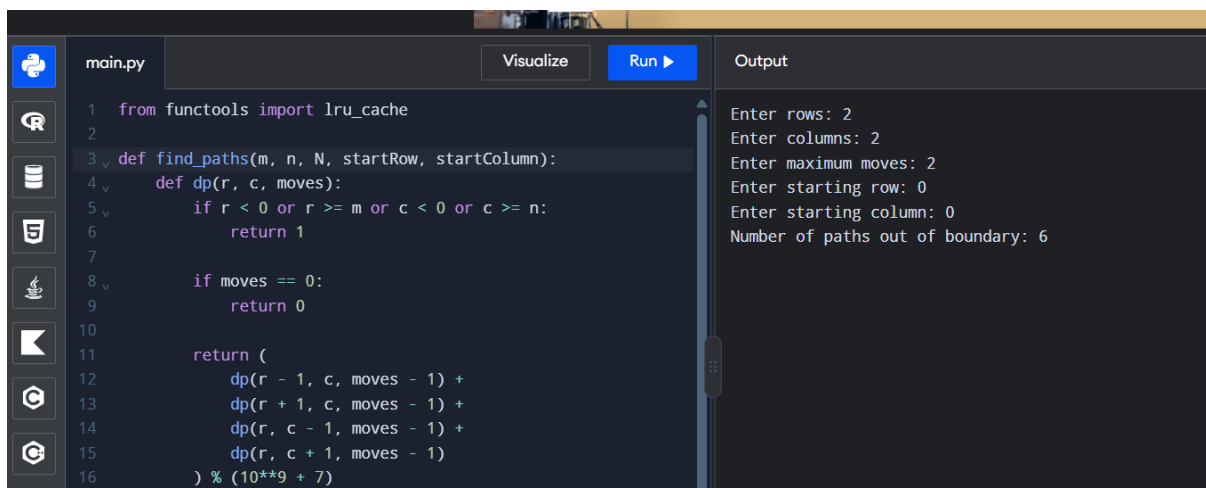
To determine the number of possible ways a ball can move outside the boundary of a given grid within a specified number of moves.

Algorithm

1. Accept the number of rows m , columns n , maximum number of moves N , and the initial position of the ball.
2. From the current cell, allow the ball to move in four directions: up, down, left, and right.
3. Check whether the ball has moved outside the grid boundary.
4. If the ball goes outside the grid, count that movement as one valid path.
5. If the maximum number of moves is reached while the ball remains inside the grid, return zero.
6. Add the results obtained from all directions and display the total number of paths.
7. Apply modulo $10^9 + 7$ to handle large values efficiently.

Procedure

1. Open Python IDLE, VS Code, or any suitable online Python compiler.
2. Create a new Python file and enter the given program.
3. Enter the number of rows, columns, maximum moves, and starting position when prompted.
4. The program processes all possible movements of the ball.
5. It counts the paths through which the ball moves outside the grid.
6. Display and record the calculated number of valid paths.



```
main.py Visualize Run ▶ Output
1 from functools import lru_cache
2
3 def find_paths(m, n, N, startRow, startColumn):
4     def dp(r, c, moves):
5         if r < 0 or r >= m or c < 0 or c >= n:
6             return 1
7
8         if moves == 0:
9             return 0
10
11        return (
12            dp(r - 1, c, moves - 1) +
13            dp(r + 1, c, moves - 1) +
14            dp(r, c - 1, moves - 1) +
15            dp(r, c + 1, moves - 1)
16        ) % (10**9 + 7)
```

Enter rows: 2
Enter columns: 2
Enter maximum moves: 2
Enter starting row: 0
Enter starting column: 0
Number of paths out of boundary: 6

Result

Thus, the number of paths through which the ball moves out of the grid boundary was successfully calculated using Python.

